# Tung Nguyen

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## Professional Experience

**Data Engineer**, Collins Aerospace (RTX) Jan 2023 - Present

- Led data engineering initiatives for Supply Chain; designed and managed 200+ Airflow, Snowflake, & Postgres pipelines supporting an organization of 600+ business users.
- Served as the team's data architect for SAP ERP-related initiatives; consolidated siloed datasets into a medallion architecture, turning high-effort, ad-hoc analyses into a scalable, analytics-ready data foundation.
- Developed a centralized contracts metadata database, enabling the company to identify $100M+ per year in assertion opportunities, with $15M of assertions sent out in the first year of operation.
- Developed an extensible simulation framework for financial analyses; integrated external economic data APIs, internal business logic, and statistical sampling methods to forecast part pricing, eliminating multiple hours of manual work per week, per analyst.
- Built an automated ML pipeline for predicting on-time delivery with daily inference and automated monthly retraining; improved existing model precision from 60% to 71% (+11%), resulting in more accurate supplier risk forecasts.
- Modernized Supply Chain's self-service data platform from an on-prem deployment to a Terraform-managed, distributed architecture using AWS ECS, RDS, DynamoDB, and S3.
- Refactored the team's Airflow deployment from a monolithic deployment to a distributed deployment with independently auto-scaled components using AWS ECS, RDS, and SQS.

**Structural Analysis Engineer**, Collins Aerospace (RTX) Jun 2018 - Jan 2023

- Designed and analyzed aerospace structures across a wide range of commercial and military platforms.
- Established structural optimization standards for the company; increasing typical weight margins from 1-5% to 10-20%; recognized as Engineer of the Year semi-finalist.
- Built multiple automation tools supporting analysis, optimization, and collaboration; introduced NLP-based methods for solving problems related to unstructured engineering data.

## Projects

**Agentic Anomaly Detection System**

- Built an end-to-end anomaly detection system for multivariate time-series data by combining forecasting algorithms with language models for interpretability.
- Automated context engineering using LangChain tool bindings and a vector database for RAG; deployed system via a lightweight chatbot interface built in Streamlit for interactive anomaly analyses.

## Education

Georgia Institute of Technology, MS in Analytics, GPA: 3.9/4.0 2025

University of California - San Diego, MS in Structural Engineering, GPA: 3.9/4.0 2018

University of California - San Diego, BS in Structural Engineering, GPA: 3.8/4.0 2017

## Technical Skills

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|---|---|---|---|
| **Programming:** | Python, Docker, Visual Basic, R, C# | **Cloud**: | AWS, Terraform |
| **Data & ETL:** | Airflow, SQL, Spark, Databricks | **Visualization**: | Tableau, PowerBI, Python |